

Prevalence and risk factors of low bone mineral density in patients with multiple sclerosis

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Objectives: To determine the prevalence and risk factors of low bone mineral density (BMD) in patients with multiple sclerosis (MS).

Methods: Patients with MS who had undergone a BMD evaluation via dual-energy X-ray absorptiometry (DXA) between January 2010 and December 2013 were included in the study. Descriptive data, BMD values, and risk factors for osteoporosis along with the details regarding MS, such as age at onset, duration of disease, clinical type of MS, expanded disability status scale (EDSS) scores, and lifetime steroid intake were obtained from the medical record database and telephone interview.

Results: The study group comprised 67 patients with a mean age of 41.1 ± 10.2 years. Of the patients, 20.9% revealed low BMD for chronological age. Vitamin D insufficiency ($25(\text{OH})\text{D} < 20 \text{ ng/ml}$) rate was 86.6%. Comorbidity and EDSS scores of patients with low BMD were significantly higher than those of the remaining patients ($P=0.000$ and $P=0.015$, respectively). Neck BMD was inversely correlated with comorbidity score, disease duration, relapse number, and lifetime steroid intake ($r=-0.270$, $r=-0.263$, $r=-0.359$, and $r=-0.314$, respectively).

Conclusion: The current study revealed that low BMD and vitamin D insufficiency were common in patients with MS. Longer disease duration, higher comorbidity score, and severe disability level led to lower BMD values. In conclusion, it is of paramount importance for clinicians to pay more attention on bone health in MS and to tailor preventative measures meticulously.

Keywords: Bone mineral density, Low bone mineral density, Multiple sclerosis, Osteoporosis

Introduction

Multiple sclerosis (MS) is a chronic inflammatory disorder of the brain and the spinal cord, which is the second leading cause of disability in young population.^{1,2} The majority of patients survives till late adulthood with long disease durations, thus experiences the inevitable consequences of MS such as osteoporosis.²

Osteoporosis is characterized by low bone mass and deterioration of the microarchitecture of bone tissue, which in turn leads to an increase in bone fragility and tendency to fractures.³ The risk of osteopenia and osteoporosis in patients with MS exceeds that in age-matched controls.⁴ The risk factors for osteoporosis in patients with MS are physical inactivity, the widespread use of glucocorticoids, vitamin D deficiency due to decreased sunlight exposure, use of anticonvulsants, and the predominance of female gender among patients with MS.²

Patients with MS are prone to bone fractures. This is in part due to impaired balance and coordination, poor vision, and side effects of the drugs such as benzodiazepines, anticonvulsants, and antidepressants, which increase the risk of falling. Osteoporosis, which is common in patients with MS, serves as one of the main risk factors for fragility fractures.⁵

The current study was conducted in order to determine the prevalence of low bone mineral density (BMD) and evaluate its relation with risk factors in a cohort of patients with MS. The overall study objective was to establish the most prominent determinants of bone health in patients with MS in order to prevent bone loss and fractures.

Materials and Methods

The present study was conducted at Cukurova University Hospital, serving mainly the patients on the southeast of Turkey. The inclusion criteria of the study group were as follows: (1) visiting neurology and/or physical medicine and rehabilitation department with a diagnosis of MS, (2) having BMD

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evaluation via dual-energy X-ray absorptiometry (DXA) between January 2010 and December 2013, and (3) aged ≥ 18 and < 65 years. The patients who had ever used any anti-osteoporotic drugs including bisphosphonates, strontium ranelate, denosumab, and teriparatide were excluded from the study.

The diagnosis of MS was based upon the McDonald criteria,⁶ and patients were categorised into four subtypes including relapsing-remitting, primary-progressive, secondary-progressive, and progressive-relapsing MS according to the consensus paper by Lublin and Reingold.⁷ After retrospective reviewing of the patients, age (year), sex, body weight (kg), height (cm), clinical type of MS, disease duration (year), expanded disability status scale (EDSS) score,⁸ lifetime grammes of high-dose methylprednisolone intake (g), number and grade of any vertebral fractures on plain thoracic and lumbar radiographs, serum levels of bone turnover markers [osteocalcin (ng/ml) and C-telopeptide (CTX) (ng/ml) (both measured via electrochemiluminescence immunoassay method)], and level of 25(OH)D (ng/ml) (measured by using high performance liquid chromatography method) were obtained from the medical electronic record database. In addition, BMD (g/cm^2) values, *T*-scores, and *Z*-scores of L1–L4 lumbar vertebra and left femoral neck sites were recorded from DXA (Hologic) scanning records. Recommendations of International Society for Clinical Densitometry were used to define ‘low BMD’.⁹ Accordingly, patients with *Z*-scores of -2.0 or below were defined as ‘below the expected range for age’ or ‘low BMD for chronological age’, whilst those with *Z*-scores above -2.0 were defined as ‘within the expected range for age’ – that is to say ‘normal BMD for chronological age’.^{9,10} Assessment of fall risk, daily calcium intake, and risk factors for low bone mass and fracture¹¹ (history of fracture, parents’ history of hip fracture, current smoking, and alcohol intake) was performed by telephone interview.

Statistical analysis of the study was performed via SPSS version 17.0 for Windows (SPSS Inc., Chicago, IL, USA). Descriptive data were stated as mean values with standard deviation. Results with *P* values less than 0.05 were considered as ‘statistically significant’. The normality of the variables was checked. Independent samples *t*-test or Mann–Whitney *U* test were used to analyze between group differences. Chi-square test was performed in order to test the differences in categorical variables. One way ANOVA test was used for multiple comparisons. The relationship between BMD values and clinical variables was evaluated by Spearman’s correlation analysis. The strength of relationships was classified as no–very weak, weak–moderate, moderate–strong, and strong correlation when the correlation coefficient (*r*) values are 0–0.25, 0.25–0.50, 0.50–0.75, and 0.75–1.00, respectively.

The ethical approval of this study was obtained from The Local Ethics Committee of Cukurova University Faculty of Medicine.

Results

The study population consisted of 67 patients (47 female, 20 male) with MS. Of the participants, 41.8% ($n=28$), 23.9% ($n=16$), and 34.3% ($n=23$) were diagnosed with MS types of relapsing-remitting, primary-progressive, and secondary-progressive, respectively. Demographics and clinical characteristics of the study population were given in Table 1. Frequencies of risk factors for low BMD and fracture were given in Table 2.

Of the study population, 20.9% revealed low BMD ($Z\text{-score} \leq -2.0$). The comparison of clinical and demographic parameters between patients with low and normal BMD was given in Table 3. Accordingly, comorbidity score and EDSS score were significantly higher in patients with low BMD than the remaining ($P=0.000$ and $P=0.015$, respectively). Additionally, there was statistically significant difference between the two groups with respect to gender ($P=0.000$).

When we compare BMD values of patients according to the subtypes of MS, lumbar BMD was significantly different among groups ($P=0.022$). A *post hoc* analysis was carried out, and the results showed that patients with secondary-progressive MS had the lowest BMD values. The lifetime steroid dosage of patients with secondary-progressive MS (29.6 ± 13.3) was higher than those with primary-progressive (23.1 ± 17.7) and relapsing-remitting MS (25.3 ± 17.8). However, the difference among groups did not reach statistical significance ($P=0.462$).

Table 1 The characteristics of the study population

Age (year)	41.1 $\pm$ 10.2
Sex (female/male)*	47 (70.1)/20 (29.9)
BMI (kg/m^2)	25.3 $\pm$ 5.0
Disease duration (year)	8.9 $\pm$ 5.8
Number of relapse	4.7 $\pm$ 3.0
Lifetime steroid intake (g)	26.1 $\pm$ 16.2
EDSS score	3.9 $\pm$ 2.0
Comorbidity score	4.4 $\pm$ 2.6
BMD (g/cm^2)	
L1–L4	0.887 $\pm$ 0.154
Femoral neck	0.742 $\pm$ 0.099
Total hip	0.789 $\pm$ 0.124
<i>T</i> -score	
L1–L4	–1.5 $\pm$ 1.2
Femoral neck	–1.2 $\pm$ 0.9
Total hip	–1.3 $\pm$ 1.0
<i>Z</i> -score	
L1–L4	–1.1 $\pm$ 1.2
Femoral neck	–0.7 $\pm$ 0.9
Total hip	–1.1 $\pm$ 1.1
25(OH)D (ng/ml)	13.7 $\pm$ 7.6
Osteocalcin (0–31.2 ng/ml)	16.5 $\pm$ 11.4
CTX (0.01–0.584 ng/ml)	0.408 $\pm$ 0.267

BMI: body mass index; EDSS: expanded disability status scale; BMD: bone mineral density; CTX: C-telopeptide.

Note: Results are expressed as mean $\pm$ SD. *Number (%).

Table 2 The risk factors for osteoporosis and fracture

Risk factor	n (%)
Falls in previous 12 months	44 (65.7)
Family history of osteoporosis	23 (34.3)
History of fracture	11 (16.4)
Parents' history of hip fracture	6 (9.0)
Current smoking	15 (22.4)
Alcohol intake	0 (0)
Glucocorticoids in the prior 6 months	54 (80.6)
Rheumatoid arthritis	0 (0)
Daily calcium intake <500 mg	42 (62.7)
Severe disability (EDSS > 5.5)	22 (32.8)
Anticonvulsants	26 (38.8)

EDSS: expanded disability status scale.

When the correlation between BMD values and clinical parameters was evaluated, the results revealed that lumbar BMD and hip BMD were negatively correlated and the number of fractures was positively correlated with EDSS score ($r = -0.259$, $r = -0.269$, and $r = 0.283$, respectively). Neck BMD was negatively correlated with comorbidity score, disease duration, relapse number, and lifetime steroid dosage ($r = -0.270$, $r = -0.263$, $r = -0.359$ and $r = -0.314$, respectively). Besides, lumbar BMD was negatively correlated with disease duration and comorbidity score ($r = -0.269$ and $r = -0.470$).

Of the patients, 47.8% ($n=32$) were using assistive device, whilst 52.2% ($n=35$) could ambulate independently. In addition, 14.9% ($n=10$) of the patients were wheelchair-bound. In order to assess the effect of the disability on BMD and bone markers, patients were categorized into two groups in accordance with their EDSS scores. Lumbar BMD of seriously disabled patients (EDSS > 5.5) was significantly lower than that of the remaining (EDSS ≤ 5.5) ($P=0.043$). However, there was no difference between these groups in terms of their bone turnover markers, vitamin D levels, total hip, and femoral neck BMD values. Regarding the BMD values in all three sites and bone turnover markers, there was no difference between patients receiving immunomodulatory therapy and those taking no immunomodulatory agents.

As means of evaluating the relation between vitamin D and bone health in MS, the study population

was divided into 'vitamin D sufficient' and 'vitamin D insufficient' groups, according to the cut-off value of 20 ng/ml for 25(OH)D. Of the patients, 86.6% was vitamin D insufficient. A comparison with regard to the BMD values and clinical parameters including disease duration, relapse number, lifetime steroid dosage, EDSS score, and number of fractures was performed between these two groups. No significant differences were detected in the above-mentioned parameters between groups.

Discussion

The aim of the present study was to determine the prevalence and risk factors of low BMD among a cohort of patients with MS. According to the results derived from this study, approximately a quarter of the MS population (20.9%) living in the southeastern part of Turkey experienced low BMD ($Z\text{-score} \leq -2.0$) and patients with secondary-progressive MS had the lowest BMD values. This prevalence seems higher when compared to that of patients without MS. For instance, a study from Turkey revealed that none of the individuals from the control group had osteoporotic values ($T\text{-score} \leq -2.5$) and only 26.8% had $T\text{-score}$ between -1.0 and -2.5 . Besides, the BMD values were significantly lower in patients with MS versus controls.¹² Fracture risk is also known to be high in MS population. According to a 2-year follow-up study, fractures without any major trauma had occurred in 22% of patients with MS versus 2% of controls.¹³ In the current study, previous fracture in adult life occurring spontaneously or following minor trauma was reported by 16.4% of the patients with MS. Moreover, more than half of the patients reported at least one fall during the previous 12 months. Therefore, it could be considered that the susceptibility to fractures in MS population is in part due to the increased fall risk with relation to the impairments in sight and balance, and partly to the reduction in BMD.¹

The risk factors for low BMD and fracture are in a wide range from gender to the dietary habits. In the

Table 3 Comparison of the patients with normal and low BMD for chronological age

	Normal BMD ($n=53$)	Low BMD ($n=14$)	P value
Age (year)	41.0 ± 10.0	41.5 ± 11.2	0.877
Sex (female/male)*	43 (81.1)/10 (18.9)	4 (28.6)/10 (71.4)	0.000
BMI (kg/m ²)	25.7 ± 5.4	23.9 ± 3.3	0.337
Disease duration (year)	8.7 ± 5.8	9.5 ± 5.7	0.626
Number of relapse	4.6 ± 2.9	5.1 ± 3.5	0.602
Lifetime steroid intake (g)	26.7 ± 17.1	23.9 ± 12.7	0.570
EDSS score	3.6 ± 1.9	5.1 ± 2.0	0.015
Comorbidity score	3.8 ± 2.2	6.9 ± 2.8	0.000
25(OH)D (ng/ml)	14.0 ± 7.7	12.7 ± 7.6	0.592
Osteocalcin (0–31.2 ng/ml)	13.9 ± 8.6	21.0 ± 14.6	0.114
CTX (0.01–0.584 ng/ml)	0.425 ± 0.276	0.375 ± 0.264	0.683

BMI: body mass index; EDSS: expanded disability status scale; BMD: bone mineral density; CTX: C-telopeptide.

Note: Results are expressed as mean ± SD.*Number (%).

present study, both lumbar and femoral neck BMD was negatively correlated with disease duration and comorbidity score. This finding is not surprising, since longer disease duration is an important determinant of impairments and comorbidities in MS population.² Although longer disease duration is a contributor for considerable consequences to health in patients with MS, low bone mass can also occur early in MS. This is suggested to be related with shared etiopathogenetic mechanisms in osteoporosis and MS.^{5,14} Therefore, preventive strategies in order to improve bone strength and to reduce fall risk should be tailored in all patients with MS, regardless of disease duration.¹⁵

Steroids play an important role in the reduction of BMD in patients with MS. Of the patients, 80.6% had a history of receiving oral/intravenous glucocorticoids in the prior 6 months. In addition, the mean lifetime steroid dosage of the study population was 26.1 ± 16.2 g. Although, a negative correlation was found between femoral neck BMD and lifetime steroid intake, there was no difference between patients with low BMD and the remaining, regarding the dosage of lifetime pulse steroids. Although prolonged steroid treatment is a distinguished contributor of low bone mass, this is not the case for high doses of methylprednisolone pulses, even in patients with MS followed up for almost 8 years.¹⁶ Moreover, decreased BMD due to high doses of pulse steroids is reversible and followed by rapid bone formation, after the cessation of steroid regimen.^{17–19}

Reduced physical activity is a well-known risk factor for osteoporosis. The increased fracture risk in patients with MS is also disability-dependent.²⁰ In the present study, it was determined that the number of vertebral fractures was increasing with relation to the increase in EDSS score. In addition, seriously disabled patients ($EDSS > 5.5$) had lower lumbar BMD values than patients who could ambulate without aid ($EDSS \leq 5.5$). Likewise, Tüzün *et al.*²¹ found that trochanteric BMD was significantly lower in non-ambulatory patients ($EDSS \geq 7.0$) than that in patients ambulating without aid ($EDSS \leq 5.5$). On the other hand, Sioka *et al.*²² evaluated BMD in ambulatory patients ($EDSS < 6.0$) with MS. They found that BMD of males was lower than sex-, age-, and BMI-matched controls, at all sites. Besides, such an association was found at the total hip site in women. Since ambulatory patients with MS are also at high risk of reduced BMD, there might be disease-associated factors other than disability that could contribute to low BMD.

In the present study, mean vitamin D level of the whole study group was below the sufficiency level and only three of the patients had an optimal vitamin D status ($25(OH)D \geq 30$ ng/ml). However, we found

no difference in BMD values between vitamin D sufficient and insufficient patients. Similarly, Triantafyllou *et al.*²³ also failed to find an association between $25(OH)D$ levels and BMD measurements. This finding along with ours argues with the previously documented correlations between low vitamin D levels and impaired BMD.^{12,24}

In conclusion, low bone mass and vitamin D insufficiency are frequent in patients with MS, particularly those with serious disabilities and longer disease durations. Therefore, physicians should be aware of the potential risks on bone health in patients with MS, in order to provide preventative measures in time.

Disclaimer Statements

Contributors All authors have contributed significantly and equally to the article.

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Conflicts of interest The authors declare no conflicts of interest.

Ethics approval The ethical approval of this study was obtained from The Local Ethics Committee of Cukurova University Faculty of Medicine.

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